

Research Title : Applications of Internet of Things (IoT) Technology in
Developing an Automated Cricket Breeding Tank for Farmer
Groups in Ban Nong Yai, Nong Yai Subdistrict, Satuek District,
Buriram Province

Researcher : Prem Enkvetchakul, Kittikoon Boonkate, Korakoch Silpakob

Organization : Faculty of Science, Buriram Rajabhat University

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ABSTRACT

This research aimed to (1) create knowledge on the application of Internet of Things (IoT) technology for controlling environmental conditions in cricket farming, (2) design and develop an automatic cricket farming cabinet capable of controlling temperature and humidity appropriately, (3) transfer the developed technology and knowledge to local farmers, and (4) evaluate the system's efficiency and user satisfaction. The study was conducted using a Research and Development (R&D) approach combined with Participatory Action Research (PAR) in Nong Kab Village, Nong Yai Subdistrict, Satuek District, Buriram Province. The system was developed using a NodeMCU ESP8266 microcontroller connected with a DHT11 sensor to measure temperature and humidity inside the cabinet. It controlled the fan and infrared lamp via a relay module and connected to the Internet for real-time monitoring through the Blynk application on smartphones. The results revealed that the developed system could automatically maintain temperature and humidity within the optimal range (25–35°C), effectively reducing labor and monitoring time. Evaluation by three experts showed that the system's overall efficiency was at a good level ($\bar{x} = 4.47$, S.D. = 0.54). Additionally, 50 farmers who participated in the technology transfer activity expressed an overall good level of satisfaction ($\bar{x} = 4.49$, S.D. = 0.49), particularly in terms of benefits such as time reduction, cost savings, and

improved production quality. The farmers also agreed that the system was convenient, practical, and could be expanded for use within the wider community.

Keywords : Internet of Things, Automated Cricket Breeding, Smart Farm